

The **.277 Fury** or **6.8×51mm Common Cartridge**,^{[4][5]} (designated as the **.277 SIG Fury** by the SAAMI^[1]) is a centerfire rimless bottlenecked rifle cartridge announced by SIG Sauer in late 2019.^[2] Its hybrid three-piece cartridge case has a steel case head and brass body connected by an aluminum locking washer to support the high chamber pressure of 80,000 psi (551.6 MPa).^[2]

Background

The cartridge was designed by SIG Sauer for the United States Army's Next Generation Squad Weapon Program (NGSW).^{[2][6]} It is dimensionally similar to the 7.62×51mm NATO service cartridge.

In December 2019, the cartridge was announced for non-military usage, along with the SIG Sauer CROSS bolt-action rifle. As a short-action rifle cartridge (cartridges having an overall length of 2.750 inches (69.85 mm) or less), increased internal ballistic performance for its cartridge case volume is accomplished by applying high gas pressure.^[7] In 2020, it was accepted by the Sporting Arms and Ammunition Manufacturers' Institute (SAAMI) as a new cartridge and chambering.^[8] In 2022 SIG Sauer announced that it intends to commercially chamber the SIG MCX Spear semi-automatic rifle in .277 Fury.^[9]

Specifications

The cartridge uses a case that is the same length and diameter as the .308 Winchester.^[10] Each "hybrid" cartridge case consists of a stainless steel base coupled to a brass body via an aluminium locking washer.^{[11][12]} Stainless steel has a significantly higher yield strength than brass, allowing the engineers to use higher maximum average pressure (MAP) chamber pressure levels.^[13] Bullets are either 135 grains (8.75 g) "match grade" with a G1 ballistic coefficient of ≈ .488 or 140 grains (9.07 g) "hunter tipped" with a G1 ballistic coefficient of ≈ .508 (ballistic coefficients are somewhat debatable).^{[12][14][15]}

These projectile ballistic coefficients are equal to the publicized specifications of the Sierra .277 135 grains (8.75 g) HPBT MatchKing and .277 140 grains (9.07 g) TGK GameKing projectiles.^[16] The SAAMI warns that MAP levels greater than 65,000 psi (448.2 MPa) may present an increased risk of unsafe

.277 Fury	
Type	Centerfire rifle
Place of origin	United States of America
Production history	
Designed	2019
Manufacturer	<u>SIG Sauer</u>
Produced	2019–present
Specifications	
Case type	<u>Rimless</u> , bottleneck
Bullet diameter	.2780 in (7.06 mm)
Land diameter	.2700 in (6.86 mm)
Neck diameter	.3100 in (7.87 mm)
Shoulder diameter	.4611 in (11.71 mm)
Base diameter	.4703 in (11.95 mm)
Rim diameter	.4720 in (11.99 mm)
Rim thickness	.0540 in (1.37 mm)
Case length	2.015 in (51.2 mm)
Overall length	2.825 in (71.8 mm)
Rifling twist	1 in 7 in. (177.8 mm)
Maximum pressure (SAAMI)	80,000 psi (550 MPa)
Ballistic performance	

cartridge case or firearm rupture and thus require cartridge case and firearm designs that depart from traditional practices (materials, construction, and other design criteria).^{[1][17][18]}

The .277 Fury SAAMI (voluntary) MAP chamber pressure of 80,000 psi (551.6 MPa) enables a 135 grains (8.7 g) projectile muzzle velocity of 3,000 feet per second (914 m/s) from a 16-inch (406 mm) barrel.^{[1][10]} It also means that the .277 Fury is normally chambered in small arms that are capable of handling the accompanying bolt thrust safely.

Commercially, reduced-power .277 Fury Elite ball FMJ ammunition using non-hybrid conventional cartridge cases is offered, that fires a 135 grains (8.7 g) projectile with a G1 ballistic coefficient of $\approx .475$ at a muzzle velocity of 2,750 feet per second (838 m/s) from a 16-inch (406 mm) barrel.^[3]

Bullet mass/type	Velocity	Energy
135 gr (9 g)	3,000 ft/s	2,694 ft·lbf
Hybrid match (910 m/s)		(3,653 J)
135 gr (9 g)	2,750 ft/s	2,267 ft·lbf
Elite ball FMJ (840 m/s)		(3,074 J)
140 gr (9 g)	2,950 ft/s	2,706 ft·lbf
Hunting (900 m/s)		(3,669 J)
Test barrel length: 16 inch (406 mm)		
Source(s): ^{[1][2][3]}		

Performance

SIG Sauer claims that the cartridge has performance superior to the 6.5mm Creedmoor, exhibiting 6 to 9 feet (1.8 to 2.7 m) less bullet drop at 1,000 yards (914 m), while delivering 20–25% greater energy.^[12] Reduced-power .277 Fury ammunition, which does not employ the stainless steel case head (.277 Fury Elite ball FMJ), and thus cannot achieve pressure above current SAAMI brass specifications, is virtually identical to the commonly available 7mm-08 Remington, which was developed in 1980.^[1]

The bullet diameter of the .277 Fury cartridge is close to that of the 7mm-08 Remington, the difference in diameter is 0.006 inch (0.15 mm) or approximately the thickness of two human hairs.^{[1][19]} The muzzle velocity and downrange energy of the reduced-power .277 Fury without stainless steel case head and 7mm-08 Remington are practically identical in similarly weighted bullets with similar ballistic coefficients.^[20]

Reduced-power .277 Fury ammunition (.277 Fury Elite ball FMJ) performs close to the brass-cased .270/308 Winchester wildcat rifle cartridge, that has a case capacity of 3.5 ml (54 grains) H₂O and the bullet, diameter-wise, fills a position between the .260 Remington and 7mm-08 Remington rifle cartridges.

Military award and designation as 6.8 common cartridge

In January 2019, the United States military began the Next Generation Squad Weapon Program to find replacements for the M4 carbine and M249 light machine gun. On April 19, 2022, the United States Army announced that it had selected SIG Sauer to build the XM7 rifle to partly replace the M4 carbine, while the XM250 automatic rifle was to replace the M249 SAW in the LMG role. In both cases it had selected the company's Fury ammunition, utilizing government provided projectiles and vendor-designed cartridges as the new weapon's ammunition.^{[21][22]}



An XM250 automatic rifle with inserted ammunition belt

The XM7 was designed to fire the 6.8×51mm SIG Fury cartridge in response to concerns that improvements in body armor would diminish the effectiveness of the 5.56×45mm NATO round used in the M4 carbine and M249 SAW and increase their lethality and effective ranges compared to common battlefield rounds such as the 5.56×45mm NATO and 7.62×51mm NATO.^{[23][24][25][26]}

Operational testing of the XM7 rifle, XM250 automatic rifle, XM157 fire control optic platform agnostic unit and the 6.8×51mm ammunition squad weaponry is expected to begin in 2024. However, this does not guarantee actual widespread future issue.^[27] The military designation for this round is *6.8 common cartridge*.^[5] The **XM1186** is the general-purpose 6.8 mm round, with other versions including reduced range rounds so weapons chambered in 6.8 mm can fire on existing ranges designed for the 5.56 mm, marking rounds for force-on-force shooting, and blank and tracer rounds.^[28]

Military solicitation

As of March 2022, the United States Army has a solicitation ongoing for an M240 machine gun 6.8×51mm conversion kit for the 7.62×51mm NATO-chambered M240B and M240L general-purpose machine guns.^{[27][29]}

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External links

- New for 2020: SIG Sauer .277 SIG Fury Ammunition (<https://www.youtube.com/watch?v=jUKA1mIYBZo>) via [YouTube](#) (Archive (<https://ghostarchive.org/varchive/jUKA1mIYBZo>))
 - .277 SIG Fury SAAMI Drawing (<https://saami.org/wp-content/uploads/2020/11/277-Sig-Public-Introduction-Orig-2020-11-11.pdf>) Archived (<https://web.archive.org/web/20230611133427/https://saami.org/wp-content/uploads/2020/11/277-Sig-Public-Introduction-Orig-2020-11-11.pdf>) June 11, 2023, at the [Wayback Machine](#)
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